Ankle MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Core Measurements

Measurement	Normal / Reference
Achilles tendon AP thickness	4-6 mm (measured on axial, midsubstance)
Achilles watershed zone	2-6 cm proximal to calcaneal insertion
Tibiofibular clear space	< 6 mm (on mortise radiograph/axial MRI)
Medial clear space	< 4 mm
Talar tilt angle	< 5 degrees (stress view equivalent)
Anterior drawer distance	< 3 mm (clinical correlation)
OLT sizing	Report AP x transverse x depth (mm); map to 9-zone grid (medial/central/lateral x anterior/central/posterior)
PT-to-FDL ratio	PT normally ~2x diameter of FDL on axial

How to Find It — Landmark-Based Identification

This section teaches you to systematically locate and evaluate ankle structures using fixed bony landmarks, the optimal plane/sequence for each, what normal looks like, and the mimics that trip up trainees. Work plane-by-plane: tendons and ligaments are best on **axial**, the Achilles and tibiotalar cartilage on **sagittal**, and the ligaments/osteochondral surfaces on **coronal**.

Orientation and Search Strategy

💡 PEARL **Match the structure to its plane**

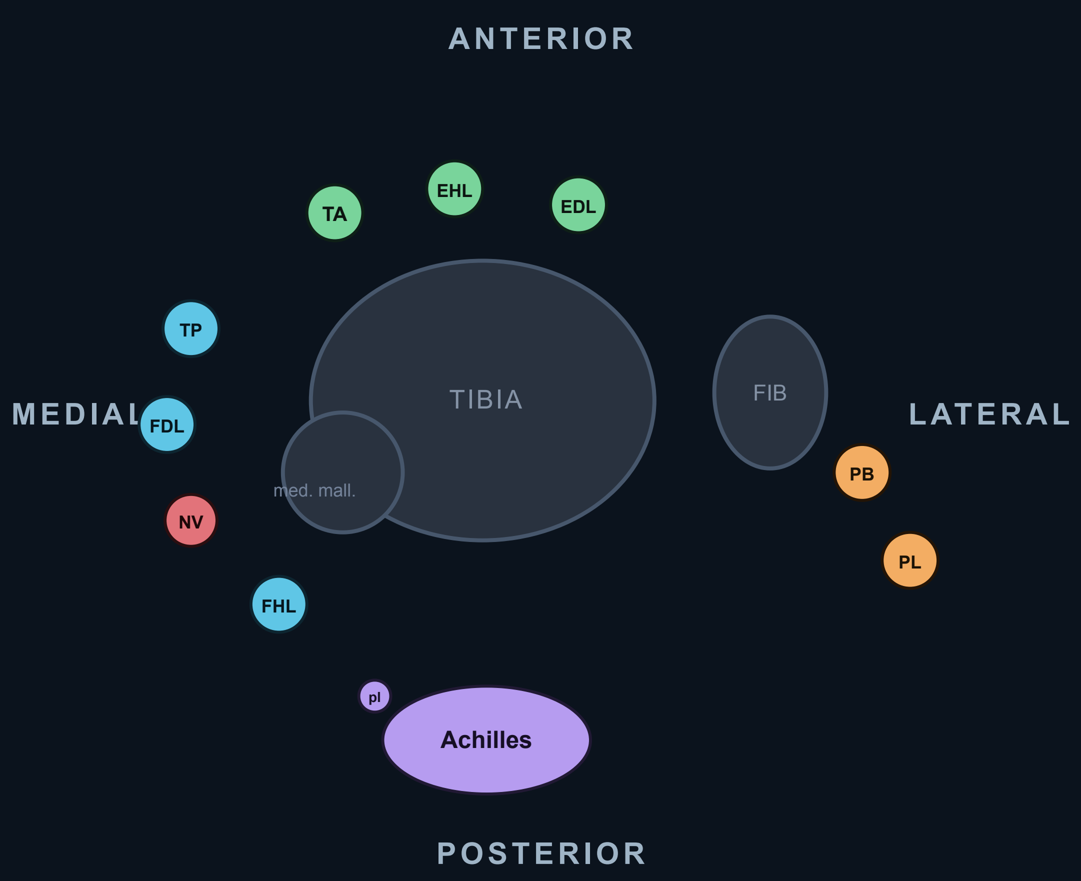
Most ankle pathology is solved by knowing which plane shows the structure best.

- **Axial:** retromalleolar tendons, all three lateral ligaments, syndesmosis, tarsal tunnel
- **Coronal:** deltoid, CFL, talar dome osteochondral surface, peroneal split
- **Sagittal:** Achilles, plantaris, tibiotalar cartilage, os trigonum/FHL, plantar fascia

📍 LANDMARK **The medial and lateral malleoli are your anchors**

Almost every tendon and ligament is named for its relationship to a malleolus.

- The **medial malleolus** anchors the deltoid and the Tom-Dick-Harry tendons.
- The **lateral malleolus (distal fibula)** anchors ATFL, CFL, PTFL, and the peroneal groove.



- **Anterior** — TA · EHL · EDL (anterior to the tibia)
- **Medial** (post. to med. malleolus) — **Tom**·TP, **Dick**·FDL, **aNd**·NV, **Harry**·FHL
- **Lateral** (post. to fibula) — peroneus Brevis (deep, on bone) · Longus (superficial)
- **Posterior** — Achilles tendon (+ plantaris, anteromedial border)

Ankle tendons in cross-section (axial). Four groups wrap the joint: **anterior** extensors, **medial** flexors behind the medial malleolus (*Tom, Dick, aNd Harry*), **lateral** peroneals behind the fibula, and the **posterior** Achilles. The neurovascular bundle (NV) sits between FDL and FHL.

Medial Retromalleolar Tendons (Axial)

🔍 HOW TO FIND IT **Posterior tibial tendon (PTT)**

On **axial** at the level of the medial malleolus, find the most **anteromedial** tendon hugging the back of the medial malleolus; it is normally ~**2×** the caliber of the adjacent FDL.

- Order behind the medial malleolus, front to back (Tom, Dick, AND Harry): **T**ibialis posterior, **F**lexor **D**igitorum longus, **and** the neurovascular bundle, **H**allucis longus.
- Follow the PTT distally to its broad insertion on the **navicular tuberosity**; the main slip there is the dominant attachment.

🔍 HOW TO FIND IT **Flexor digitorum longus (FDL)**

Immediately **posterior to the PTT** on axial, smaller (about half the PTT diameter). It crosses superficial to FHL at the **knot of Henry** in the midfoot.

🔍 HOW TO FIND IT **Flexor hallucis longus (FHL)**

The **most posterolateral** of the medial tendons, lying in its groove **between the talar tubercles**. On sagittal it runs deep, hooking under the sustentaculum tali toward the great toe.

- The **neurovascular bundle** (posterior tibial artery/vein and tibial nerve) sits between FDL and FHL — this is the "and" in Tom, Dick, AND Harry.

⚠️ PITFALL **Magic-angle on the curving PTT**

Where the PTT curves around the medial malleolus it may show intermediate signal on short-TE sequences (**magic angle**, fibers near 55° to B0). Confirm on a long-TE (T2/PD) sequence before calling tendinosis or a tear.

⚠️ PITFALL **A small fluid rim is normal around the PTT/FHL**

A thin sheath of fluid is physiologic. The **FHL sheath communicates with the tibiotalar joint** in many people, so isolated FHL fluid is often just a joint effusion, not tenosynovitis. Demand circumferential fluid plus a thickened/enhancing sheath before calling true tenosynovitis.

📏 MEASURE **Calling PTT dysfunction**

Compare the PTT to the FDL on axial. A PTT that is **enlarged, thinned, or equal to/smaller than the FDL**, or shows longitudinal splits/intratendinous T2 signal, indicates posterior tibial tendon dysfunction (PTTD).

Lateral Peroneal Tendons (Axial)

🔍 HOW TO FIND IT **Peroneus brevis and longus in the retromalleolar groove**

On **axial** at the lateral malleolus, find the two peroneals in the **retromalleolar (fibular) groove**. The **brevis is anterior/medial and sits directly against the fibula**; the **longus is posterolateral** to it.

- Mnemonic: **brevis is bony** (against bone), longus is lateral/superficial.

⚠️ PITFALL **The brevis boomerang/C-shape split sign**

A normal brevis is round/oval. When it becomes **flattened, C-shaped or boomerang-shaped wrapping around the longus**, suspect a **longitudinal split tear** of the peroneus brevis — classically associated with a shallow/convex fibular groove and a chronically lax superior peroneal retinaculum.

📍 LANDMARK **Peroneal tubercle and cuboid tunnel for the longus**

Follow the **longus** distally: it passes below the **peroneal tubercle** of the lateral calcaneus, then turns medially through the **cuboid tunnel/groove** beneath the cuboid toward the first metatarsal base.

- An **os peroneum** (sesamoid in the longus at the cuboid) is a normal variant; check it isn't fractured/diastatic if the longus is symptomatic.

⚠️ PITFALL **Superior peroneal retinaculum and subluxation**

The **superior peroneal retinaculum (SPR)** holds the tendons in the groove. A torn/stripped SPR allows the tendons to **sublux anterolaterally over the fibular tip** — look for an empty or shallow groove and a retinacular avulsion fleck.

Lateral Collateral Ligaments (Axial; CFL also Coronal)

🔍 HOW TO FIND IT **Anterior talofibular ligament (ATFL)**

On **axial** at the level of the malleolar tip, find the thin flat band running from the **anterior margin of the lateral malleolus to the talar neck/body**. It is the most anterior of the three.

- It is the **first and most commonly torn** ligament in an inversion sprain.

🔍 HOW TO FIND IT **Calcaneofibular ligament (CFL)**

Find it on **axial or coronal** running from the **lateral malleolar tip obliquely down to the lateral calcaneus**, passing **deep to the peroneal tendons**. Use the peroneals as your signpost — the CFL lies directly under them.

- A CFL tear is suggested when fluid tracks into the **peroneal tendon sheath** because the ligament forms part of its floor.

🔍 HOW TO FIND IT **Posterior talofibular ligament (PTFL)**

On **axial**, the thick, **striated/fan-shaped** band from the **malleolar fossa (medial fibula) to the posterolateral talus**. Its normal striated appearance is not a tear.

💡 PEARL **Lateral sprain order: front to back**

Ligaments fail in sequence with inversion: **ATFL first, then CFL, then PTFL**. An isolated PTFL tear without ATFL/CFL injury is rare — re-check your reading if you see it.

Syndesmosis (Axial)

🔍 HOW TO FIND IT **AITFL and PITFL**

On **axial just above the plafond**, the **anterior inferior tibiofibular ligament (AITFL)** runs obliquely from the anterolateral tibia (Chaput tubercle) to the anterior fibula (Wagstaffe), and the **posterior inferior tibiofibular ligament (PITFL)** runs from the posterior tibia (Volkmann) to the posterior fibula. Both are normally striated.

📍 LANDMARK **Interosseous ligament and membrane**

Above the AITFL/PITFL, the **interosseous ligament** is the distal thickening of the **interosseous membrane** filling the tibiofibular space. Disruption here, especially with a high fibular fracture, signals a **high (Maisonneuve) syndesmotic injury** — always look up the leg.

📏 MEASURE **Tibiofibular clear space**

Assess the syndesmosis on **axial ~1 cm (10 mm) above the tibial plafond**. The **tibiofibular clear space (between the lateral tibia and medial fibula) should be < ~6 mm**; widening suggests syndesmotic diastasis.

Medial Ligaments (Coronal primary; Axial adjunct)

🔍 HOW TO FIND IT **Deltoid ligament — deep and superficial layers**

Best on **coronal**. The **deep deltoid** is the short, striated **tibiotalar** band deep to the tendons connecting the medial malleolus to the talus; the **superficial deltoid** is the broader fan spanning malleolus to **navicular, spring ligament, sustentaculum, and talus**.

- Deep deltoid disruption is the key finding in a "deltoid-positive" unstable ankle fracture (medial clear-space widening).

🔍 HOW TO FIND IT **Spring (superomedial calcaneonavicular) ligament**

On **axial/coronal** just deep and plantar to the PTT, find the hammock-like band from the **sustentaculum tali to the navicular** that cradles the talar head. The **superomedial component** lies immediately deep to the PTT.

💡 PEARL **Adult acquired flatfoot triad**

PTT dysfunction rarely travels alone. Check all three: **(1) PTT tear/tenosynovitis, (2) spring ligament insufficiency, (3) sinus tarsi changes**. Finding one should prompt a hunt for the others.

Sinus Tarsi and Tarsal Tunnel

📍 LANDMARK **Sinus tarsi contents**

On **sagittal/coronal**, the **sinus tarsi** is the fat-filled cone between the talus and calcaneus, lateral to the subtalar joint. It contains the **interosseous talocalcaneal ligament** (medial/deep) and the **cervical ligament** (anterolateral).

- **Normal = bright fat.** Replacement of that fat by T1-dark, T2-bright fluid/scar = **sinus tarsi syndrome** (often partnered with PTT/spring disease).

🔍 HOW TO FIND IT **Tarsal tunnel**

On **axial behind the medial malleolus**, the tunnel is roofed by the **flexor retinaculum** and contains (front to back) PTT, FDL, the **posterior tibial neurovascular bundle**, and FHL. Trace the **tibial nerve** as it divides into medial and lateral plantar branches; a ganglion or accessory muscle here causes **tarsal tunnel syndrome**.

Achilles Tendon and Plantaris (Sagittal + Axial)

🔍 HOW TO FIND IT **Achilles tendon**

Best on **sagittal**; confirm on **axial**. Follow it from the musculotendinous junction to the **posterosuperior calcaneal insertion**. The normal anterior margin is **flat or slightly concave** on axial — a **convex/rounded anterior margin** means thickening.

📏 MEASURE **Achilles thickness and watershed zone**

Normal **AP thickness ~4–6 mm** on sagittal. The **hypovascular watershed zone is ~2–6 cm above the insertion** — the classic site of degenerative tears. Insertional disease instead centers at the calcaneus with retrocalcaneal bursitis and a Haglund deformity.

⚠️ PITFALL **Don't mistake plantaris for an intact Achilles**

The **plantaris tendon** runs along the **anteromedial border of the Achilles**. In an Achilles rupture an intact plantaris can be misread as residual intact fibers — trace it separately on axial; it is a thin discrete band medial to the main tendon.

💡 PEARL **Sheath vs paratenon — name the inflammation correctly**

The **Achilles tendon has NO true (synovial) tendon sheath** — like the patellar and quadriceps tendons, it is invested only by a **paratenon** (loose areolar connective tissue). So peritendinous inflammation here is **paratenonitis** (synonyms: **peritendinitis**, **paratendinitis**) — **never** "tenosynovitis."

- **Tenosynovitis** is reserved for the **sheathed** ankle tendons: tibialis posterior, FDL, FHL, peroneus brevis/longus, tibialis anterior, EHL, EDL.
- **Paratenonitis** (Achilles) → fluid/edema in the **paratenon**, classically a thin rim **anterior to the tendon within Kager fat** (and circumferentially on axial), with **enhancement**. The tendon itself may be normal or thickened.
- When tendinosis coexists, call it **paratenonitis with tendinosis** (part of the "Achilles tendinopathy" spectrum).

🔍 HOW TO FIND IT **Achilles paratenonitis (peritendinitis)**

On **axial**, look for a **circumferential bright (T2/STIR) rim of fluid/edema around the Achilles**, often most conspicuous along the **anterior surface in the Kager fat pad**; on **sagittal**, a thin peritendinous fluid line hugging the front and back of the tendon.

- Pure paratenonitis: peritendinous fluid/enhancement with a **normal-signal, normal-caliber tendon**.
- Distinguish from **retrocalcaneal bursitis** (focal fluid in the bursa **between the tendon and the posterosuperior calcaneus**) — a discrete bursa, not a peritendinous rim.

Talar Dome Osteochondral Lesions (Coronal + Sagittal)

📍 LANDMARK **The 9-zone grid**

Mentally divide the talar dome into a **3×3 grid** (medial/central/lateral × anterior/middle/posterior) on coronal/sagittal to standardize lesion location.

	MEDIAL	CENTRAL	LATERAL
ANTERIOR			
MIDDLE			traumatic
POSTERIOR	atraumatic (most common)		

medial dome → deeper / cup-shaped · lateral dome → shallower / wafer

Talar dome 9-zone grid. Map every osteochondral lesion to one of the nine cells. Classic locations: **medial-posterior** (atraumatic, deeper) and **lateral-central** (traumatic, shallower).

💡 PEARL **Medial-posterior vs lateral-central**

Two classic locations: **medial dome lesions are posterior, deeper, cup-shaped, and often atraumatic**; **lateral dome lesions are central/anterior, shallower, wafer-shaped, and usually traumatic**.

📏 MEASURE **Calling instability**

An osteochondral fragment is **unstable** when **high T2 fluid signal fully undermines the fragment**, there are subchondral **cysts**, a displaced fragment, or a >5 mm lesion. A bright fluid line completely encircling the base = loose/unstable.

Posterior Impingement and Accessory Ossicles (Sagittal/Axial)

🔍 HOW TO FIND IT **Os trigonum / Stieda process**

On **sagittal**, find the bony ossicle or elongated process at the **posterolateral talus**, just lateral to the FHL groove. Marrow/soft-tissue edema around it in a plantarflexion athlete (dancer, soccer) indicates **posterior ankle impingement**.

- The **FHL tendon runs immediately medial** to the os trigonum/Stieda process; FHL tenosynovitis frequently accompanies posterior impingement.

🔍 HOW TO FIND IT **Anterolateral gutter impingement**

On **axial**, the **anterolateral gutter** sits between the tibia/fibula and talus anterior to the ATFL. Post-sprain **scar/synovial soft-tissue masses (meniscoid lesion)** filling this gutter cause anterolateral impingement; look for low-signal soft tissue replacing the normal fat.

Plantar Fascia (Sagittal)

🔍 HOW TO FIND IT **Plantar fascia origin**

On **sagittal**, find the thin low-signal band arising from the **medial calcaneal tuberosity** and running forward along the sole. Normal **proximal thickness is < ~4 mm**; **focal thickening >4 mm with adjacent edema** at the calcaneal origin indicates plantar fasciitis.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartment] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

> **F:** Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

> **I:** No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

> **F:** Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

> **I:** Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

> **F:** Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

> **I:** Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

> **F:** Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

> **I:** Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

> **F:** Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

> **I:** Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Ankle MRI Tells

- Achilles: watershed zone 2-6 cm above insertion; concave anterior margin lost = tendinopathy; plantaris may mimic intact longitudinal fibers in complete rupture
- PTT: PT should be 2x diameter of FDL on axial; when equal or smaller = chronic dysfunction; always check spring ligament and hindfoot alignment together
- Peroneals: C-shape/boomerang on axial = brevis split; look for retinaculum integrity at the tip of the fibula; longus pathology peaks at the cuboid tunnel
- Sinus tarsi: fat signal loss = sinus tarsi syndrome, often sequela of inversion injury; check cervical and interosseous ligaments
- OLT: medial posterior (atraumatic) vs lateral central (traumatic); fluid undermining fragment = unstable
- Spring ligament: always check with PTT dysfunction (adult acquired flatfoot triad: PTT + spring + deltoid)
- Anterolateral gutter: soft tissue thickening after inversion sprain = anterolateral impingement (meniscoid lesion)
- Os trigonum: marrow edema in os trigonum + FHL tenosynovitis = posterior impingement in dancers/plantarflexion athletes

Normal / Negative Templates

Normal Achilles

> **F:** Achilles tendon demonstrates normal caliber and uniform low T1 and T2 signal with intact concave anterior margin. AP thickness measures [X] mm. No fusiform thickening, intrasubstance signal abnormality, or peritendinous fluid. Retrocalcaneal bursa is not distended. Calcaneal insertion is unremarkable without enthesophyte.

I: Normal Achilles tendon. No tendinopathy, tear, or retrocalcaneal bursitis.

Normal ankle

F: Lateral ligament complex including the anterior talofibular, calcaneofibular, and posterior talofibular ligaments is intact without thickening or signal abnormality. Syndesmosis including the anterior inferior tibiofibular ligament, posterior inferior tibiofibular ligament, and interosseous membrane is intact. Tibiofibular clear space is normal. Deep and superficial deltoid ligament complex and spring ligament are intact. Achilles tendon is normal in caliber and signal without tendinosis, tear, or paratenonitis. The posterior tibial tendon, flexor digitorum longus tendon, flexor hallucis longus tendon, peroneus brevis and longus tendons, and anterior compartment tendons are intact without tendinosis, tear, or tenosynovitis. Talar dome articular cartilage is preserved without osteochondral defect or subchondral marrow abnormality. No ankle joint effusion. Plantar fascia is normal in thickness and signal. Sinus tarsi fat signal is preserved. Osseous structures demonstrate normal marrow signal without fracture or stress reaction.

I: Normal ankle MRI. No ligamentous, tendinous, chondral, or osseous abnormality.

Useful Caveats

- Accessory soleus muscle: normal variant in Kager fat pad; may mimic soft tissue mass on axial images
- Os trigonum vs Stieda process: os trigonum is a separate ossicle (synchondrosis with talus), Stieda process is an elongated lateral tubercle of the posterior talus; both can cause posterior impingement
- Peroneal subluxation: dynamic phenomenon, static MRI may underestimate; look for flat/convex retromalleolar groove and retinaculum disruption
- Magic angle effect: tendons coursing at 55 degrees to B0 (especially peroneal tendons at retromalleolar groove) may show artifactual intermediate signal on short TE sequences
- Low-lying peroneus brevis muscle belly: normal variant that predisposes to crowding and brevis split tears
- Accessory peroneus quartus: present in ~10-20% of individuals; courses posteromedial to peroneals

Achilles Tendon

Normal Achilles

F: Achilles tendon demonstrates normal caliber and uniform low T1 and T2 signal with intact concave anterior margin. AP thickness measures [X] mm. No fusiform thickening, intrasubstance signal abnormality, or peritendinous fluid.

I: Normal Achilles tendon without tendinopathy or tear.

Achilles tendinosis (watershed zone)

F: Fusiform thickening of the Achilles tendon at the watershed zone approximately [X] cm proximal to the calcaneal insertion with AP diameter measuring [X] mm. There is intermediate T2 intrasubstance signal with loss of the normal concave anterior margin. No discrete fluid-signal cleft or fiber discontinuity. Peritendinous soft tissues are [normal/mildly edematous].

I: Achilles tendinosis at the watershed zone without tear.

Achilles partial-thickness tear

> **F:** Focal partial-thickness thinning of the Achilles tendon [X] cm proximal to the calcaneal insertion with fluid-signal cleft involving the [anterior/medial/lateral] fibers and extending approximately [X]% of the tendon cross-section. Remaining intact fibers are [thickened with tendinosis/normal]. Surrounding peritendinous edema is present.

> **I:** Partial-thickness Achilles tendon tear at the watershed zone involving approximately [X]% of the tendon cross-section with background tendinosis.

Achilles complete rupture

> **F:** Complete discontinuity of the Achilles tendon [X] cm proximal to the calcaneal insertion with [Y] cm retraction gap filled with hemorrhage and edema. Proximal tendon stump demonstrates wavy/retracted morphology. Distal stump is [thickened/frayed]. Plantaris tendon is [intact and interposed within the gap/also torn]. Surrounding Kager fat pad demonstrates edema and hemorrhagic signal.

> **I:** Complete Achilles tendon rupture with [Y] cm gap. Plantaris tendon is [intact/torn].

Achilles paratenonitis (peritendinitis)

> **F:** Peritendinous edema and fluid surrounding the Achilles tendon, most pronounced along the anterior surface within the Kager fat pad and circumferential on axial images, with [peritendinous enhancement]. The Achilles tendon itself is [normal in caliber and signal / mildly thickened with background tendinosis]. No intrasubstance fluid-signal cleft or fiber discontinuity. (Note: the Achilles has no synovial sheath, so this represents paratenonitis rather than tenosynovitis.)

> **I:** Achilles paratenonitis (peritendinitis) [with mild tendinosis]. No tendon tear.

Insertional Achilles tendinopathy

> **F:** Thickening and intermediate T2 signal of the Achilles tendon at the calcaneal insertion with [enthesophyte/posterior calcaneal spur] at the insertion site. [Intratendinous calcification.] Posterosuperior calcaneal prominence (Haglund morphology) is [present/absent]. Retrocalcaneal bursa is [distended with fluid/not distended].

> **I:** Insertional Achilles tendinopathy with [enthesophyte] [and Haglund deformity] [and retrocalcaneal bursitis].

Retrocalcaneal bursitis (without tendon abnormality)

> **F:** Distended retrocalcaneal bursa with fluid signal interposed between the Achilles tendon and posterosuperior calcaneus. Achilles tendon is normal in caliber and signal without tendinosis or tear. Posterosuperior calcaneal prominence is [present/absent].

> **I:** Retrocalcaneal bursitis without Achilles tendinopathy. [Haglund morphology noted.]

Tibialis Posterior

Normal posterior tibial tendon

> **F:** Posterior tibial tendon demonstrates normal caliber with low T1 and T2 signal and ovoid cross-section posterior to the medial malleolus. PT diameter is approximately twice that of the adjacent flexor digitorum longus tendon. No tendon sheath fluid or peritendinous edema.

> **I:** Normal posterior tibial tendon.

PTT tenosynovitis without tendon signal abnormality

> **F:** Fluid within the posterior tibial tendon sheath surrounding an otherwise normal-caliber tendon with preserved low signal and ovoid cross-section. No intrasubstance signal abnormality or tendon thickening. PT-to-FDL diameter ratio is preserved.

> **I:** Posterior tibial tenosynovitis without tendinosis or tear.

PTT tendinosis with loss of cross-sectional morphology

> **F:** Posterior tibial tendon is thickened with intermediate T2 intrasubstance signal and loss of the normal ovoid cross-section posterior to the medial malleolus. PT-to-FDL diameter ratio is [reduced/approximately 1:1]. No discrete longitudinal split or full-thickness discontinuity. [Mild tenosynovitis.]

> **I:** Posterior tibial tendinosis [with tenosynovitis]. PT-to-FDL ratio is abnormally [reduced], suggesting [early/progressive] posterior tibial tendon dysfunction.

PTT longitudinal split tear

> **F:** Longitudinal split of the posterior tibial tendon at/distal to the medial malleolus with fluid signal interposed between tendon fragments within a distended tendon sheath. Tendon cross-section demonstrates a bilobed or split morphology. [Spring ligament is intact/attenuated.] [Hindfoot alignment is normal/valgus.]

> **I:** Longitudinal split tear of the posterior tibial tendon [with tenosynovitis]. [Spring ligament attenuation.] [Hindfoot valgus.]

PTT chronic dysfunction / advanced disease

> **F:** Marked thinning and attenuation of the posterior tibial tendon, which is [equal to/thinner than] the adjacent flexor digitorum longus tendon. Tendon demonstrates diffuse intermediate signal with loss of normal morphology. Spring ligament is [thickened/attenuated/disrupted]. [Hindfoot valgus deformity is present.] [Sinus tarsi fat signal is partially effaced.]

> **I:** Chronic posterior tibial tendon dysfunction with advanced tendon attenuation. [Spring ligament insufficiency.] [Hindfoot valgus deformity consistent with adult acquired flatfoot.]

Peroneal Tendons

Normal peroneal tendons

> **F:** Peroneus brevis and longus tendons are normal in caliber and signal within the retromalleolar groove. Superior peroneal retinaculum is intact. No subluxation, tenosynovitis, or tendon signal abnormality. Peroneus longus courses normally through the cuboid tunnel.

> **I:** Normal peroneal tendons without tear, subluxation, or tenosynovitis.

Peroneal tenosynovitis

> **F:** Fluid within the common peroneal tendon sheath surrounding otherwise normal-caliber peroneus brevis and longus tendons. No intrasubstance tendon signal abnormality, split tear, or subluxation. Superior peroneal retinaculum is intact.

> **I:** Peroneal tenosynovitis without tendon tear or subluxation.

Peroneus brevis split tear

> **F:** Longitudinal split tear of the peroneus brevis tendon at/distal to the retromalleolar groove with C-shaped/boomerang tendon morphology on axial images. Peroneus longus tendon is interposed between the brevis fragments. [Peroneal tenosynovitis.] [Low-lying brevis muscle belly contributing to retromalleolar crowding.] Superior peroneal retinaculum is [intact/attenuated].

> **I:** Longitudinal split tear of the peroneus brevis tendon [with peroneal tenosynovitis]. [Low-lying muscle belly.]

Peroneal subluxation / retinaculum disruption

> **F:** Peroneus brevis [and longus] tendon is subluxed anteriorly from the retromalleolar groove with disruption/stripping of the superior peroneal retinaculum from the distal fibula. [Small cortical flake avulsion from the lateral malleolar ridge.] Retromalleolar groove is [shallow/flat/convex]. [Peroneal tenosynovitis.]

> **I:** Peroneal tendon subluxation with superior peroneal retinaculum disruption [and fibular rim avulsion].

Peroneus longus tendinosis at cuboid tunnel

> **F:** Peroneus longus tendon is thickened with intermediate T2 intrasubstance signal at the cuboid tunnel/peroneal sulcus of the cuboid. No discrete tendon discontinuity. [Os peroneum is intact/fragmented.]

> **I:** Peroneus longus tendinosis at the cuboid tunnel. [Os peroneum is intact/fragmented.]

Lateral Ligaments

Normal lateral ligament complex

> **F:** Anterior talofibular ligament, calcaneofibular ligament, and posterior talofibular ligament are intact with normal low signal and expected morphology. No periligamentous edema or soft tissue swelling.

> **I:** Intact lateral ankle ligament complex.

ATFL sprain (grade 1-2)

> **F:** Anterior talofibular ligament demonstrates edema and thickening with preserved fiber continuity. No complete discontinuity or fluid-filled gap. [Mild anterolateral soft tissue edema.]

> **I:** Grade [1/2] ATFL sprain with preserved ligament continuity.

ATFL partial tear with fiber disruption

> **F:** Anterior talofibular ligament is thickened and edematous with partial fiber disruption and irregularity at the [fibular/talar] attachment. Some intact fibers are identified. Anterolateral soft tissue edema is present. CFL is [intact/also injured].

> **I:** High-grade partial ATFL tear. [CFL is intact/also injured.]

ATFL complete tear

> **F:** Complete discontinuity of the anterior talofibular ligament with fluid-filled gap between the fibular and talar attachments. Anterolateral soft tissue edema and hemorrhage. [Calcaneofibular ligament is intact/also torn.]

> **I:** Complete ATFL tear. [CFL is intact/also torn.]

Combined ATFL + CFL tear

> **F:** Complete discontinuity of the anterior talofibular ligament with fluid-filled gap. Calcaneofibular ligament is also discontinuous/absent with surrounding edema. Posterior talofibular ligament is [intact/injured]. [Talar dome subchondral marrow edema at the [medial/lateral] shoulder.] [Anterolateral gutter soft tissue thickening.]

> **I:** Complete tear of the ATFL and CFL. [Talar dome contusion/osteochondral injury.] [PTFL is intact/injured.]

Syndesmosis

Normal syndesmosis

> **F:** Anterior inferior tibiofibular ligament, posterior inferior tibiofibular ligament, and interosseous membrane are intact with normal low signal. Tibiofibular clear space is normal. No fluid in the interosseous recess above the level of the tibial plafond.

> **I:** Intact syndesmosis.

AITFL sprain / partial disruption

> **F:** Anterior inferior tibiofibular ligament demonstrates thickening and edema without complete fiber discontinuity. Interosseous membrane is [intact/thickened]. Posterior inferior tibiofibular ligament is intact. Tibiofibular clear space measures [X] mm. [Mild fluid in the interosseous recess.]

> **I:** AITFL sprain without complete disruption. [Mild widening of the interosseous recess.]

AITFL tear with widened clear space

> **F:** Complete discontinuity of the anterior inferior tibiofibular ligament with fluid-filled gap. Tibiofibular clear space measures [X] mm (widened). Fluid extends into the interosseous recess above the tibial plafond. Interosseous membrane demonstrates [edema/partial disruption]. Posterior inferior tibiofibular ligament is [intact/injured].

> **I:** High ankle sprain with complete AITFL tear and widened tibiofibular clear space. [Interosseous membrane injury.] [PITFL is intact/injured.]

Complete syndesmotic disruption

> **F:** Complete discontinuity of the anterior inferior tibiofibular ligament, posterior inferior tibiofibular ligament, and interosseous membrane with frank widening of the tibiofibular clear space measuring [X] mm. Fluid fills the interosseous recess. [Medial clear space widening measuring X mm.] [Deltoid ligament injury.]

> **I:** Complete syndesmotic disruption involving the AITFL, PITFL, and interosseous membrane with tibiofibular diastasis. [Medial clear space widening suggesting deltoid incompetence.]

Deltoid / Spring Ligament

Normal deltoid and spring ligament

> **F:** Deep and superficial components of the deltoid ligament are intact with normal low signal. Spring ligament (superomedial calcaneonavicular ligament) is intact with normal thickness and signal. No medial soft tissue edema.

> **I:** Intact deltoid and spring ligament complexes.

Deep deltoid partial disruption

> **F:** Deep component of the deltoid ligament (posterior tibiotalar ligament) demonstrates thickening and edema with partial fiber disruption at the [talar/tibial] attachment. Superficial deltoid fibers are [intact/also edematous]. [Medial clear space is normal/widened.]

> **I:** Deep deltoid ligament partial tear. [Superficial deltoid is intact/sprained.]

Spring ligament partial disruption

> **F:** Spring ligament (superomedial calcaneonavicular ligament) is thickened and edematous with partial fiber disruption and loss of normal taut morphology. [Posterior tibial tendon demonstrates concurrent tendinosis/tear.] [Hindfoot alignment is normal/valgus.]

> **I:** Spring ligament partial tear. [Concurrent PTT dysfunction.] [Hindfoot valgus.]

Combined deltoid + spring attenuation with hindfoot valgus

> **F:** Deep deltoid ligament is attenuated with partial fiber disruption. Spring ligament is thickened and lax with partial disruption. Posterior tibial tendon is [thinned/torn]. Hindfoot valgus deformity is present. [Sinus tarsi fat effacement.] [Talonavicular joint uncovering.]

> **I:** Combined deltoid and spring ligament insufficiency with posterior tibial tendon dysfunction and hindfoot valgus, consistent with adult acquired flatfoot deformity.

Talar Dome / OLT

Normal talar dome

> **F:** Talar dome articular cartilage is preserved in thickness and signal without focal defect, subchondral marrow edema, or cystic change. Subchondral bone plate is intact.

> **I:** Normal talar dome. No osteochondral lesion.

Subchondral edema with cartilage signal change (early OLT)

> **F:** Focal subchondral marrow edema at the [medial posterior/lateral central] talar dome with overlying cartilage signal heterogeneity but intact subchondral bone plate. No discrete osteochondral fracture line or chondral fissure. Area measures approximately [X] x [Y] mm.

> **I:** Early osteochondral change of the [medial/lateral] talar dome with subchondral edema and chondral softening. No unstable fragment.

OLT with fracture line and chondral fissuring (stable)

> **F:** Osteochondral lesion of the [medial posterior/lateral central] talar dome measuring [X] x [Y] x [Z] mm with subchondral fracture line, overlying chondral fissuring, and surrounding marrow edema. [Subchondral cystic change measuring X mm.] No fluid undermining the fragment. Subchondral bone plate is [intact/minimally depressed].

> **I:** [Stable] osteochondral lesion of the [medial/lateral] talar dome with chondral fissuring [and subchondral cystic change].

Unstable OLT with fluid undermining

F: Osteochondral lesion of the [medial/lateral] talar dome measuring [X] x [Y] x [Z] mm with fluid-signal undermining the subchondral fragment on T2-weighted images. Overlying cartilage demonstrates [fissuring/partial-thickness defect]. [Subchondral cystic change.] Fragment appears partially detached but remains in the crater.

I: Unstable osteochondral lesion of the [medial/lateral] talar dome with fluid undermining the fragment. Surgical consultation recommended.

Displaced OLT fragment

F: Empty osteochondral crater at the [medial/lateral] talar dome measuring [X] x [Y] mm with exposed subchondral bone and surrounding marrow edema. Displaced osteochondral fragment is identified [within the joint/in the anterior gutter/in the posterior recess] measuring [X] mm. [Reactive synovitis and joint effusion.]

I: Displaced osteochondral lesion of the [medial/lateral] talar dome with [intra-articular loose body].

Subchondral cyst without unstable fragment

F: Focal subchondral cyst at the [medial/lateral] talar dome measuring [X] mm with surrounding mild marrow edema. Overlying articular cartilage is [intact/thinned]. No discrete fracture line or unstable osteochondral fragment.

I: Subchondral cyst of the [medial/lateral] talar dome [with overlying chondral thinning]. No unstable osteochondral fragment.

Impingement

No anterior or posterior impingement

F: No anterior tibial or talar osteophytes. No synovial thickening in the anterior recess. Posterior talus/os trigonum region is unremarkable. Flexor hallucis longus tendon is normal in signal and caliber without tenosynovitis.

I: No anterior or posterior ankle impingement.

Anterior bony impingement

F: Anterior tibial plafond osteophyte and/or opposing talar neck osteophyte with synovial thickening and edema in the anterior ankle recess. [Bone marrow edema within the osteophytes.] [Osteophyte measures X mm.] [Small chondral defect at the anterior tibiotalar articulation.]

I: Anterior ankle impingement from tibiotalar osteophytes with anterior synovitis.

Posterior impingement (os trigonum / Stieda process)

F: [Os trigonum/prominent Stieda process (elongated lateral tubercle of the posterior talus)] with marrow edema and surrounding soft tissue edema. [Synchondrosis edema/irregularity between the os trigonum and talus.] Flexor hallucis longus tendon is [normal/demonstrates fluid in its tendon sheath/thickened with tendinosis] as it courses between the medial and lateral tubercles. [Posterior tibiotalar joint capsular thickening.]

I: Posterior ankle impingement syndrome involving the [os trigonum/Stieda process] [with FHL tenosynovitis].

Anterolateral soft tissue impingement

F: Soft tissue thickening and synovial/scar tissue in the anterolateral gutter of the ankle with adjacent capsular edema. [Anteroinferior tibiofibular ligament thickening.] [Residual ATFL thickening consistent with prior sprain.]

I: Anterolateral ankle impingement/anterolateral gutter scarring, likely sequela of prior lateral ankle sprain.

Plantar Fascia

Normal plantar fascia

F: Plantar fascia demonstrates normal thickness (approximately [X] mm) and uniform low signal at the calcaneal origin and throughout the central band. No perifascial edema or enthesophyte.

I: Normal plantar fascia.

Plantar fasciitis (proximal, central band)

F: Thickening of the proximal central band of the plantar fascia at the calcaneal origin measuring [X] mm in thickness (normal < 4 mm) with increased T2/STIR signal. Perifascial edema is present in the adjacent subcalcaneal fat pad. [Calcaneal enthesophyte/spur measuring X mm.] [Reactive calcaneal marrow edema at the insertion.]

I: Plantar fasciitis involving the proximal central band [with calcaneal enthesophyte] [and reactive calcaneal marrow edema].

Plantar fascia partial tear / focal discontinuity

F: Focal discontinuity of the plantar fascia [X] cm distal to the calcaneal origin with fluid-signal gap measuring [X] mm in the [central/medial/lateral] band. Surrounding perifascial edema and hemorrhagic signal. [Remaining fibers are intact/also thickened with fasciitis.]

I: Partial-thickness plantar fascia tear [with background plantar fasciitis].

Plantar fascia midsubstance thickening

F: Fusiform thickening and intermediate T2 signal of the plantar fascia at the midsubstance/midfoot level measuring [X] mm without discrete discontinuity or calcaneal origin involvement.

I: Midsubstance plantar fascial thickening/fasciosis without tear.

Sinus Tarsi / Tarsal Tunnel

Normal sinus tarsi and tarsal tunnel

F: Sinus tarsi demonstrates normal fat signal without edema or fibrosis. Cervical and interosseous ligaments are intact. Tarsal tunnel contents are unremarkable with normal tibial nerve caliber and no mass lesion. Plantar musculature demonstrates normal signal.

I: Normal sinus tarsi and tarsal tunnel.

Sinus tarsi syndrome

F: Normal sinus tarsi fat signal is replaced by edema and/or fibrotic/scarring tissue. Cervical ligament is [thickened/disrupted]. Interosseous talocalcaneal ligament is [thickened/disrupted]. [Associated lateral ligament injury.] [Subtalar joint effusion/synovitis.]

I: Sinus tarsi syndrome with fat effacement and [cervical/interosseous] ligament [thickening/disruption], likely sequela of prior inversion injury.

Tarsal tunnel - mass with nerve compression

F: [Ganglion cyst/schwannoma/accessory muscle/varicosities] within the tarsal tunnel measuring [X] cm with mass effect upon the tibial nerve. Tibial nerve appears [compressed/displaced]. [Denervation edema in the plantar musculature on fluid-sensitive sequences.]

I: Tarsal tunnel [mass type] with compression of the tibial nerve. [Denervation signal in the plantar muscles.]

Tarsal tunnel - edema with nerve enlargement

F: Edema and soft tissue thickening within the tarsal tunnel. Tibial nerve is enlarged measuring [X] mm (normal < 5 mm) with increased T2 signal. [Flexor retinaculum is thickened.] [Abnormal signal in the abductor hallucis/abductor digiti minimi/flexor digitorum brevis consistent with denervation edema.]

I: Tarsal tunnel syndrome with tibial nerve enlargement [and denervation changes in the plantar musculature].

Effusion / Bone / Other

Trace ankle effusion

F: Trace physiologic fluid in the tibiotalar joint. No synovial thickening or intra-articular body.

I: Trace ankle joint effusion, likely physiologic.

Small effusion with synovial thickening

F: Small ankle joint effusion with mild synovial thickening. No intra-articular loose body.

I: Small ankle joint effusion with mild synovitis.

Moderate effusion

F: Moderate ankle joint effusion distending the anterior and posterior recesses. [Synovial thickening.] [Intra-articular body in the posterior recess measuring X mm.]

I: Moderate ankle joint effusion [with synovitis]. [Intra-articular loose body.]

Calcaneal stress reaction / fracture

F: [Focal/diffuse] marrow edema within the calcaneal body/tuberosity on fluid-sensitive sequences [with/without] discrete linear low-signal fracture line on T1-weighted images. [Periosteal edema along the calcaneal cortex.] No displaced fracture fragment.

I: Calcaneal [stress reaction/stress fracture] [without/with visible fracture line].

Subcutaneous soft tissue edema

> **F:** Subcutaneous soft tissue edema along the [medial/lateral/posterior] ankle without organized fluid collection or abscess. [Associated with adjacent ligamentous/tendinous injury.]

> **I:** Subcutaneous edema of the [medial/lateral/posterior] ankle, reactive.

Search Pattern

- **1. Check history, indication, and priors.**
- **2. Assess study adequacy, technique, and limitations.**
- **3. Look at localizers** for incidental pathology.
- **4. Check for effusion/collection** on fluid-weighted sequences: most will be at tibiotalar joint; cysts/collections.
- **5. Examine bony cortex and marrow space:**
- Cortical breaks/fractures, osteophytes, loose bodies, heterotopic ossification.
- Talar dome (osteochondral lesions); marrow replacement; hematopoietic marrow.
- Accessory ossicles/variants (especially Type 1-3 os naviculare).
- **6. Assess tendons:**
- Continuity, position/size, signal on PD and fluid-sensitive; check for accessory muscles/tendons (adjacent to flexors, peroneals, soleus).
- Posterior: Achilles (concave up and completely dark on axial fluid-sensitive; check retrocalcaneal and pre-Achilles bursas; fluid around tendon = paratenonitis).
- Medial (T-D-H: tibialis posterior [should be 2x size of other medial tendons], flexor digitorum longus, flexor hallucis longus; axials best along superior extent near medial malleolus; small fluid physiologic; check FDL/FHL crossing in foot for edema/avulsion).
- Lateral (peroneus brevis and longus; small fluid physiologic; check for tear or subluxation; assess retinaculum for thickening/disruption).
- Anterior (T-H-D: tibialis anterior, extensor hallucis longus, extensor digitorum longus; check for abnormal location, thickening, signal; assess retinaculum).
- **7. Assess ligaments:**
- Lateral: anterior and posterior tibiofibular (syndesmotic) ligaments (fluid-sensitive axials above tibiotalar articulation, should be thin and dark); anterior and posterior talofibular ligaments (below tibiotalar articulation, thin and dark); calcaneofibular ligament (coronals through ankle, deep to peroneals).
- Medial: deltoid (normal triangular shape spanning tibia, calcaneus, navicular — identify at least deep tibiotalar and superficial tibiospring portions).
- Spring (plantar calcaneonavicular) ligament (deep to deltoid, just under talar head — check especially if posterior tibialis dysfunction/pes planus; identify at least superomedial calcaneonavicular portion from sustentaculum tali to medial navicular, deep to posterior tibialis).
- Incidental: Lisfranc ligament (spanning medial cuneiform and 2nd metatarsal base, check for disruption on short axis views).
- **8. Assess plantar fascia:** thickening, perifascial edema, disruption.
- **9. Assess hindfoot joints:**

- Tibiotalar joint (PD and fluid-sensitive; articular cartilage loss; osteochondral lesions; effusion).
- Subtalar (talocalcaneal) joint (coalition at medial and posterior facets; sinus tarsi fat signal/edema).
- Midfoot joints and tarsometatarsal (Lisfranc) joints if visualized (edema, injury, erosion, degeneration).
- **10. Assess subcutaneous tissues and musculature:** T1 precontrast, fluid-sensitive; atrophy, edema, inflammation, mass; note accessory muscles; neurovascular bundle at medial ankle.
- **11. Double check** marrow signal, muscles, and subcutaneous tissues.
- **12. Proofread report.**

Infection Search Pattern

- **1. Marrow/bone changes:**
- Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
- Marrow edema (isolated finding raises concern in diabetics/high-risk).
- Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
- Rim-enhancing collection = abscess.
- Superficial lesion without enhancement = ulcer.
- Enhancement along fascial planes = infectious/inflammatory fasciitis.
- Non-enhancement of normal structures = ischemia.
- Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
- Joint effusion with marrow replacement on both sides of joint.
- Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
- Sinus tract connecting abnormal marrow signal to skin surface.
- Disappearance of subchondral cysts on sequential imaging.
- Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Source Notes

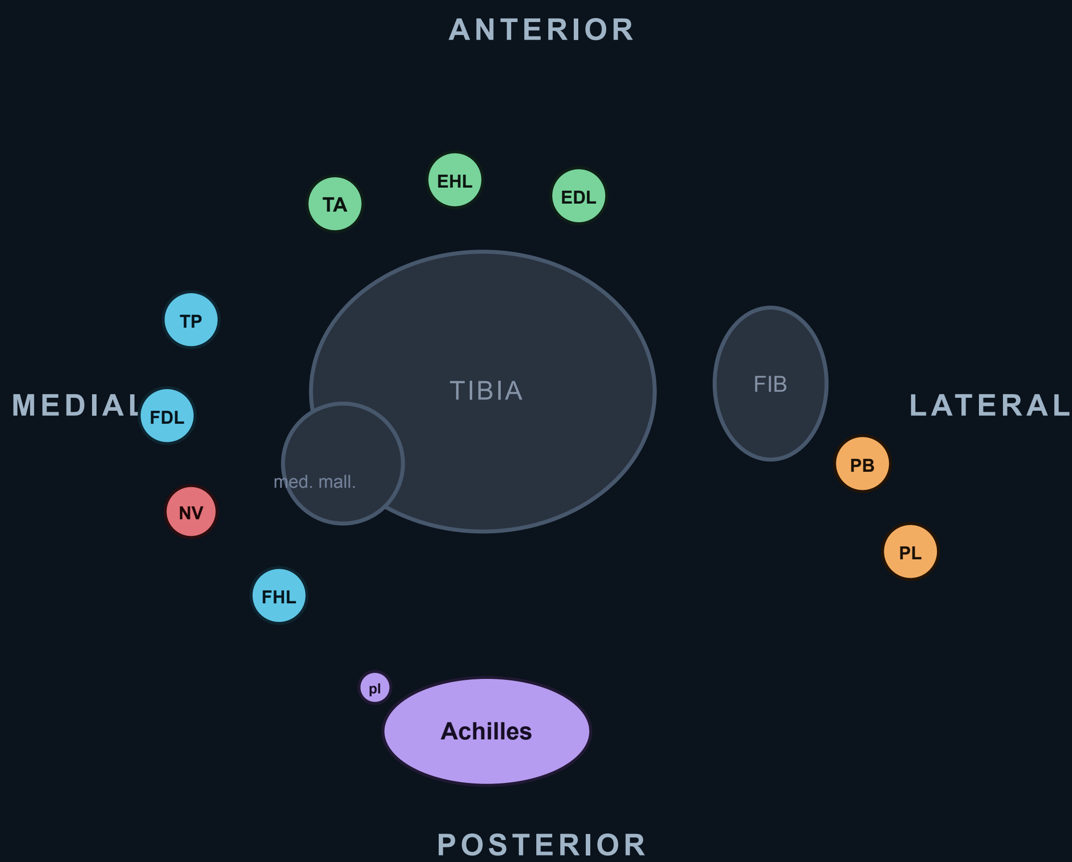
Built as a practical dictation aid from standard MSK MRI reporting patterns, review-level radiology references, and the existing shoulder cheat-sheet style. PowerScribe template selections for Achilles, PTT, peroneals, lateral ligaments, syndesmosis, deltoid/spring, talar dome OLT, impingement, plantar fascia, sinus tarsi, tarsal tunnel, and effusion were incorporated as individual F/I entries. Use surgical correlation and local report conventions when needed.

Test Yourself & Visual Pearls

Work through these before you sign the report — the tendons, ligaments, and bony os variants of the ankle reward pattern recognition.

Tendons around the ankle

On an axial image at the level of the medial malleolus, the flexor tendons run in a fixed anterior-to-posterior order that you can name on sight.



- **Anterior** — TA · EHL · EDL (anterior to the tibia)
- **Medial** (post. to med. malleolus) — **Tom**·TP, **Dick**·FDL, **aNd**·NV, **Harry**·FHL
- **Lateral** (post. to fibula) — peroneus Brevis (deep, on bone) · Longus (superficial)
- **Posterior** — Achilles tendon (+ plantaris, anteromedial border)

Ankle tendons in cross-section (axial). Four groups wrap the joint: **anterior** extensors, **medial** flexors behind the medial malleolus (*Tom, Dick, aNd Harry*), **lateral** peroneals behind the fibula, and the **posterior** Achilles. The neurovascular bundle (NV) sits between FDL and FHL.

🔑 KEY POINTS **Tendon map**

- Medial (behind medial malleolus, front-to-back): **T**ibialis posterior → flexor **D**igitorum longus → posterior tibial **N**eurovascular bundle → flexor **H**allucis longus — **Tom, Dick, aNd Harry**.
- Lateral (behind lateral malleolus): peroneus **brevis** lies anteromedial (against the fibula), peroneus **longus** posterolateral.
- Achilles has no synovial sheath → it gets **paratenonitis**, not tenosynovitis; a true sheath (and sheath fluid) implies a different tendon.

? QUIZ **A patient with a flatfoot deformity has fluid and thickening of the most anterior medial tendon. Which tendon, and what is its role?**

? QUIZ **Behind the lateral malleolus, which peroneal tendon sits closer to the bone, and why does it matter?**

Talar dome lesions

Localize every osteochondral lesion on a 9-zone grid (medial/central/lateral × anterior/middle/posterior) — location predicts mechanism and the chondral surface tells you about stability.

	MEDIAL	CENTRAL	LATERAL
ANTERIOR			
MIDDLE			traumatic
POSTERIOR	atraumatic (most common)		

medial dome → deeper / cup-shaped · lateral dome → shallower / wafer

Talar dome 9-zone grid. Map every osteochondral lesion to one of the nine cells. Classic locations: **medial-posterior** (atraumatic, deeper) and **lateral-central** (traumatic, shallower).

💡 TIP **Classic dichotomy: medial-posterior lesions are deeper, cup-shaped, and often atraumatic/degenerative, whereas lateral lesions are shallow, wafer-shaped, and trauma-related. Lateral lesions are more likely to displace.**

? QUIZ **On a fluid-sensitive sequence, a talar dome osteochondral fragment has a rim of high T2 signal completely encircling its base. Stable or unstable?**

Rapid review

? QUIZ **Name the lateral collateral ligaments in order of how commonly they tear in an inversion injury.**

? QUIZ **How do you grade an ATFL injury on MRI?**

? QUIZ Which is the key anterior syndesmotic ligament, and what measurement flags a syndesmotic injury?

⚠ PITFALL Isolated AITFL tear with a normal clear space is easy to dismiss — but always trace the fibula proximally. A high fibular fracture (Maisonneuve) means the whole syndesmosis and deltoid are disrupted even if the ankle "looks" reduced.

? QUIZ A patient with chronic flatfoot has a torn medial soft-tissue sling. Name the classic flatfoot triad of structures.

? QUIZ Where does the peroneus brevis split tear typically occur, and what bony sign and restraint do you check?

? QUIZ Distinguish Achilles paratenonitis from "tenosynovitis."

? QUIZ Differentiate insertional from non-insertional Achilles disease, and name the associated bony bump.

? QUIZ What plantar fascia thickness is abnormal, and where is it measured?

? QUIZ What occupies the sinus tarsi, and what does loss of its fat signal indicate?

⚠ PITFALL Don't call every posterior talar ossicle a fracture. An os trigonum is a smooth, corticated unfused ossicle behind the talus; marrow edema across it (or in the adjacent process) signals os trigonum syndrome / posterior impingement, often in dancers and with FHL tenosynovitis.